

Introduction to Gaussian Processes

Andrew Yuan

CAMP REU @ Cal Poly Pomona

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Statistics Background

- ▶ Univariate Normal Distribution

$$y \sim \mathcal{N}(\mu, \sigma^2)$$

where μ = Mean and σ^2 = Variance

- ▶ Multivariate Normal Distribution

$$\mathbf{Y} \sim \mathcal{MVN}(\boldsymbol{\mu}, \Sigma)$$

where $\boldsymbol{\mu}$ = Mean and Σ = PSD Covariance Matrix

- ▶ The marginal and conditional distributions of a MVN are also normal
- ▶ Affine transformations of a MVN are also normal

Gaussian Processes

- ▶ **Definition:** A Gaussian Process (GP) is a collection of random variables $(f(\mathbf{x}))_{\mathbf{x} \in \mathcal{X}}$ such that for any finite collection $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathcal{X}$, the vector $\mathbf{f} := [f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)]^\top$ has a multivariate normal distribution.

$$f \sim \mathcal{GP}(\mu(x), k(x, x'))$$

- ▶ $\mu(x)$ is the mean function and $k(x, x')$ is the covariance kernel
- ▶ Assume a prior mean function of 0
- ▶ GP is typically defined by the choice of Kernel

Gaussian Processes

- ▶ Common choices are Matern and Radial Basis Function Kernel; each are parameterized by hyperparameters (e.g., lengthscale ℓ , outputscale η^2 , etc.)
- ▶ Matern Kernel:

$$k_{\text{Matern}}(x, x') = \eta^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} d \right)^\nu K_\nu \left(\sqrt{2\nu} d \right)$$

where $d = (x - x')^\top \Theta^{-2} (x - x')$, $\Theta = \text{diag}(\ell)$, and D_ν is the modified Bessel Function

- ▶ RBF Kernel:

$$k_{\text{RBF}}(x, x') = \eta^2 \exp \left(-\frac{1}{2} (x - x')^\top \Theta^{-2} (x - x') \right)$$

Gaussian Processes

Some Kernel Properties:

1. Kernels are symmetric (Covariances are symmetric)

$$K(x, z) = K(z, x)$$

2. K_1, K_2 kernels then $aK_1(x, z) + bK_2(x, z)$ is a kernel for $a, b \in \mathbb{R}^+$
3. K_1, K_2 kernels, then $K_1(x, z)K_2(x, z)$ is also a kernel
4. (2) and (3) imply that any polynomial evaluated at K_1 is a kernel

$$K(x, z) = \text{pol}^+[K_1(x, z)]$$

5. (4) implies that $K(x, z) = \exp(K_1(x, z))$ is a kernel

Gaussian Process Regression

- ▶ GP Regression is a non-parametric approach to regression problems:

$$y(\mathbf{x}) = f(\mathbf{x}) + \epsilon(\mathbf{x})$$

- ▶ The fundamental assumption is that $f(\mathbf{x})$ follows a Gaussian process; place a GP prior on $f(\mathbf{x})$
- ▶ Provide a training dataset so that the GP model can "learn" and ultimately predict $f(\mathbf{x})$
- ▶ To obtain GP model that best explains data under prior, optimize the log marginal likelihood using MLE criterion:

$$\log p(\mathbf{y} | X_{train}) = -\frac{1}{2} \mathbf{y}^\top (K_{train} + \Sigma_\epsilon)^{-1} \mathbf{y} - \frac{1}{2} \log |K_{train} + \Sigma_\epsilon|$$

Gaussian Process Regression

- To predict at a new test point $x_* \in \mathcal{X}$, compute the posterior:

$$f_*(x_*) := f(x_*) \mid \mathcal{D}_{train} \sim \mathcal{N}(m(x_*), s^2(x_*))$$

with

$$m(x_*) = K(x_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1} \mathbf{y}$$

$$s^2(x_*) = k(x_*, x_*) - K(x_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1} K(X_{train}, x_*)$$

$$K(x_*, X_{train}) = \begin{bmatrix} k(x_1, x_*) \\ \vdots \\ k(x_n, x_*) \end{bmatrix}^\top$$

Gaussian Process Regression

- ▶ Similarly, to predict multiple test points $\{\mathbf{x}_{1*}, \dots, \mathbf{x}_{m*}\} = \mathbf{X}_* \subset \mathcal{X}$, compute the posterior:

$$f_*(\mathbf{X}_*) := f(\mathbf{X}_*) \mid \mathcal{D}_{train} \sim \mathcal{MVN}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*)$$

with

$$\boldsymbol{\mu}_* = K(\mathbf{X}_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1} \mathbf{y}$$

$$\boldsymbol{\Sigma}_* = k(\mathbf{X}_*, \mathbf{X}_*) - K(\mathbf{X}_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1} K(X_{train}, \mathbf{X}_*)$$

$$K(X_{train}, \mathbf{X}_*) = \begin{bmatrix} k(x_1, x_{1*}) & \cdots & k(x_1, x_{m*}) \\ \vdots & \ddots & \vdots \\ k(x_n, x_{1*}) & \cdots & k(x_n, x_{m*}) \end{bmatrix}^\top$$

Applications of GPs

Bayesian Optimization (BO): A strategy to predict global optimum of black-box functions that are typically expensive to evaluate. Assumptions include the function being continuous and the feasible set to be bounded and low dimension.

- ▶ Strategy: Treat unknown objective function as a random function and place a prior over it
- ▶ Common choice: Use a **Gaussian Process** as the prior to model function uncertainty
- ▶ GPs provide both a predictive mean and a predictive variance

Applications of GPs

BO consists of 2 parts: a Bayesian statistical model and an acquisition function for deciding where to sample next.

A general algorithm is as follows:

1. Apply a GP prior on objective function f with the initial data points n_o in training set X_{train}
2. Assume a simulation budget of N points and some parameter space of potential points H
3. Apply an acquisition function (e.g., Expected Improvement) on H to decide the next point x_n to add onto X_{train}
4. Compute $f(x_n)$ and then refit the GP on f with X_{train}
5. Repeat the process until X_{train} reaches the budget of N

Applications of GPs

One of the most commonly used acquisition functions is **Expected Improvement (EI)**:

- ▶ $f(x_n)$ is unknown until computed; ideally want $f(x_n)$ to be "better" than $f_n^* := \max_{m \leq n} f(x_m)$
- ▶ Find expected value of this improvement and select the next point x_n in order to maximize this improvement
- ▶ EI balances exploration (high uncertainty) and exploitation (high mean prediction)

Applications of GPs

Define:

$$El_n = E_n[[f(x) - f_n^*]^+]$$

This can be evaluated to closed form via Int by Parts resulting in:

$$El_n(x) = [\Delta_n(x)]^+ + \sigma_n(x)\varphi(Z) - |\Delta_n(x)|\Phi(Z)$$

where $\Delta_n(x) = \mu_n(x) - f_n^*$ and $Z = \frac{\Delta_n(x)}{\sigma_n(x)}$

- ▶ φ is the PDF of the Gaussian Dist. and Φ is the CDF of the Gaussian Dist.
- ▶ In Step 4 of the BO algorithm, the point with the largest expected improvement x_n is evaluated (ties are broken arbitrarily).

Applications of GPs

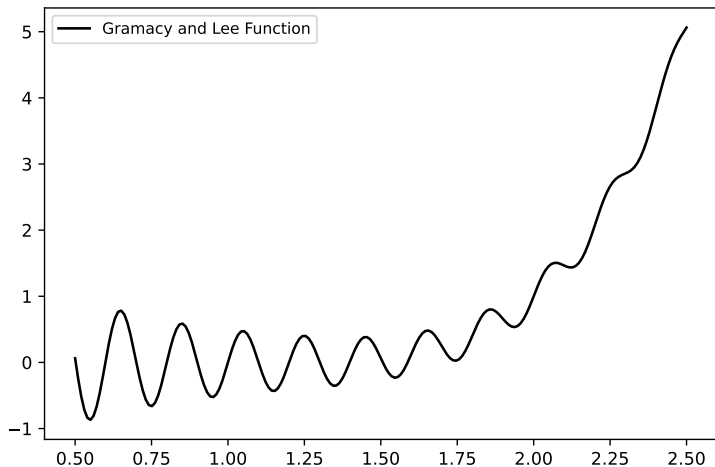


Figure: Gramacy and Lee Test Function: $f(x) = \frac{\sin(10\pi x)}{2x} + (x-1)^4$

Applications of GPs

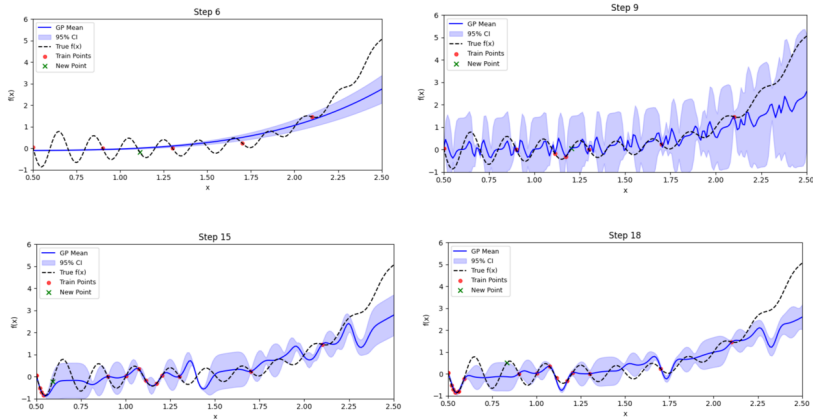


Figure: BO Algorithm: Periodic + Polynomial Kernel and EI

Applications of GPs

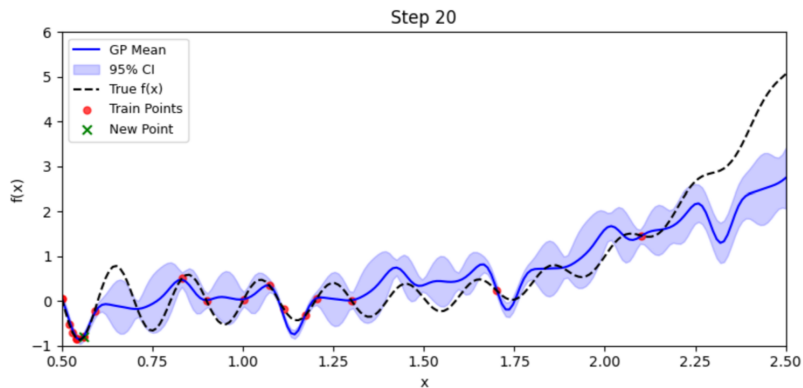


Figure: BO Algorithm: True Argmin at 0.5486 vs BO Argmin at 0.5503